

Customer Segmentation Project – Professional Analyst Insights

Executive Summary

This project performs customer segmentation on a large e-commerce dataset (250,000 transactions, 50,000 customers) using RFM (Recency, Frequency, Monetary) analysis and K-Means clustering. The goal is to identify distinct customer groups based on purchasing behavior to enable targeted marketing, improve retention, and maximize customer lifetime value (CLV).

Key Findings

1. Data Overview

- **Total transactions:** 250,000
- **Total customers:** 50,000
- **Date range:** January 2020 to September 2023
- **Average transaction value:** ₹2,725
- **Average customer age:** 44 years
- **Churn rate:** ~20% (49,874 out of 250,000 customers)
- **Return rate:** ~19% of transactions had returns

2. Sales Trends

Yearly Sales:

- 2020: ₹185.3 million
- 2021: ₹182.8 million
- 2022: ₹182.9 million
- 2023: ₹130.3 million (incomplete year, data through September)

Observation: Sales have been relatively stable from 2020 to 2022, with a slight decline in 2023 (likely due to incomplete data).

3. Customer Segmentation Results

The K-Means algorithm identified **3 distinct customer clusters** with a silhouette score of **0.377**, indicating reasonable separation between groups.

Cluster	Customer Count	Avg Recency	Avg Frequency	Avg Monetary	Churn Rate	Top Product Category
0	~16,667	Low	High	High	19.75%	Books
1	~16,667	Medium	Medium	Medium	20.08%	Clothing
2	~16,667	High	Low	Low	20.17%	Clothing

Cluster Interpretation:

- **Cluster 0 (High-Value Customers):** Recent purchasers with high frequency and high spending. These are your best customers.
- **Cluster 1 (Regular Customers):** Moderate recency, frequency, and spending. These customers show consistent but not exceptional engagement.
- **Cluster 2 (At-Risk Customers):** Low recency (haven't purchased recently), low frequency, and low monetary value. These customers are at risk of churning.

4. Churn Analysis

All three clusters have similar churn rates (~20%), which suggests that churn is not strongly correlated with RFM segments in this dataset. However, **Cluster 2** has the highest churn rate (20.17%), making it a priority for retention campaigns.

5. Product Category Preferences

- **Cluster 0:** Prefers Books (30,669 purchases)
- **Cluster 1:** Prefers Clothing (7,653 purchases)
- **Cluster 2:** Prefers Clothing (36,755 purchases)

This indicates that different customer segments have distinct product preferences, which can be leveraged for personalized recommendations.

Strategic Recommendations

For Cluster 0 (High-Value Customers)

- **Loyalty Programs:** Offer exclusive rewards, early access to new products, and personalized discounts.
- **Upselling:** Recommend higher-value products or premium versions of items they frequently purchase.
- **Referral Programs:** Encourage them to refer friends and family with incentives.
- **Priority Support:** Provide dedicated customer service to maintain their satisfaction.

For Cluster 1 (Regular Customers)

- **Cross-Selling:** Recommend complementary products based on their purchase history.
- **Bundle Offers:** Create product bundles to increase average order value.
- **Seasonal Campaigns:** Send targeted promotions during holidays or special occasions.
- **Engagement Emails:** Share product recommendations, new arrivals, and personalized content.

For Cluster 2 (At-Risk Customers)

- **Win-Back Campaigns:** Send personalized emails with limited-time discounts or special offers.
- **Re-Engagement:** Remind them of products they previously purchased or browsed.
- **Feedback Surveys:** Understand why they haven't purchased recently and address any issues.
- **Low-Cost Incentives:** Offer free shipping or small discounts to encourage a return purchase.

General Recommendations

1. **Personalized Marketing:** Use cluster assignments to tailor email campaigns, ads, and product recommendations.
2. **Monitor Churn:** Implement early warning systems to identify customers showing signs of churn (e.g., decreasing purchase frequency).

3. **Product Recommendations:** Leverage category preferences to suggest relevant products to each cluster.
4. **A/B Testing:** Test different marketing strategies for each cluster to optimize conversion rates.
5. **CLV Optimization:** Focus on increasing the lifetime value of Cluster 1 customers by moving them toward Cluster 0 behavior.

Technical Strengths

- **Large Dataset:** 250,000 transactions provide robust statistical power.
- **RFM Analysis:** Industry-standard method for customer segmentation.
- **Feature Scaling:** Proper preprocessing ensures K-Means performs optimally.
- **Silhouette Score:** Validation metric confirms reasonable cluster separation.
- **Actionable Insights:** Clear business recommendations derived from data.

Areas for Improvement

1. **Cluster Validation:** Consider using the elbow method or gap statistic to confirm the optimal number of clusters ($k=3$).
2. **Visualization:** Add PCA or t-SNE plots to visualize cluster separation in 2D/3D space.
3. **Temporal Analysis:** Analyze how clusters evolve over time (e.g., do customers move from Cluster 2 to Cluster 1?).
4. **Additional Features:** Incorporate demographic data (age, gender, location) for richer segmentation.
5. **Predictive Modeling:** Build a churn prediction model using machine learning (e.g., Random Forest, XGBoost) to proactively identify at-risk customers.
6. **Revenue Contribution:** Calculate the revenue contribution of each cluster to prioritize marketing spend.

Conclusion

This customer segmentation project successfully identifies three distinct customer groups with actionable insights for targeted marketing. By implementing the recommended strategies, the business can improve customer retention, increase average order value, and maximize lifetime value. The next steps should focus on deploying these insights into marketing campaigns and continuously monitoring cluster performance over time.

